

Impact Assessment of PM SURYA GHAR MUFT BIJLI YOJANA

Proposal to Uttar Pradesh New & Renewable Energy
Development Agency (UPNEDA)

October 2024



Quality Council of India

(An autonomous body set up by
Ministry of Commerce & Industry, Govt. of India)

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1. BACKGROUND

The Government of India has approved the PM Surya Ghar: Muft Bijli Yojana on 29th February, 2024 to increase the share of solar rooftop capacity and empower residential households to generate their own electricity. The scheme has an outlay of Rs 75,021 crore and is to be implemented till FY 2026-27. The scheme will be implemented by a National Programme Implementation Agency (NPIA) at the National level and by the State Implementation Agencies (SIAs) at the State level.

The Distribution Utility (DISCOMs or Power/Energy Departments) is the State Implementation Agencies (SIA) at the State/UT level. Under the scheme, the DISCOMs are required to put in place several facilitative measures for promotion of rooftop solar in their respective areas such as availability of net meters, timely assessment and commissioning of installations, vendor registration and management, interdepartmental convergence for solarizing government building etc.

The scheme provides for a subsidy of 60% of the solar unit cost for systems up to 2 kW capacity and 40 percent of additional system cost for systems between 2 kW to 3 kW capacity. The subsidy has been capped at 3 kW capacity.

In the state of Uttar Pradesh, this scheme is being implemented by the Uttar Pradesh New and Renewable Energy Development Agency (UPNEDA), reflecting a strategic effort to realize the state's renewable energy infrastructure and address local energy needs. It has a significant role in the rollout of the PM Surya Ghar: Muft Bijli Yojana within the state. The agency is responsible for managing the scheme's implementation, including the distribution of subsidies, monitoring of installations, and ensuring compliance with scheme guidelines. UPNEDA's efforts are integral to achieving the scheme's objectives and ensuring its success in improving energy access and promoting sustainable practices across the state.

2. ABOUT PM SURYA GHAR MUFT BIJLI YOJANA

The 'PM Surya Ghar: Muft Bijli Yojana' is a central government initiative designed to promote solar energy adoption for residential electricity needs. The scheme primarily focuses on encouraging households to install solar power systems by offering financial incentives and subsidies. The scheme supports environmental sustainability by reducing reliance on traditional energy sources and provides financial assistance to make solar installations more accessible and affordable for average households. This initiative aligns with India's broader goals of enhancing renewable energy use and achieving energy security. Additionally, the scheme supports the Uttar Pradesh Vision Solar Policy 2022, further advancing the state's solar energy objectives.

The proposed impact assessment of the 'PM Surya Ghar: Muft Bijli Yojana' to be conducted by UPNEDA in collaboration with Quality Council of India (QCI), is designed to thoroughly evaluate the scheme's effectiveness and outcomes. This assessment will explore various dimensions, including how the scheme has affected the lives of its beneficiaries. The assessment aim to provide practical recommendations to enhance the scheme's efficiency and effectiveness, ensuring better implementation across different regions of the state. Furthermore, it will address the challenges (if any) faced by stakeholders, identify necessary interventions, and propose improvements to the existing solar ecosystem in Uttar Pradesh.

3. ABOUT QUALITY COUNCIL OF INDIA (QCI)

The Quality Council of India (QCI) has been set up as a non-profit autonomous society registered under Societies Registration Act XXI of 1860 to establish an accreditation structure in the country and to spread quality movement in India by undertaking a National Quality Campaign.

QCI was established by the Government of India together with the Indian Industry represented by the three major industry associations: ASSOCHAM, CII and FICCI. The Department for Promotion of Industry and Internal Trade (DPIIT), Ministry of Commerce and Industry is the nodal point for all matters related to QCI.

The Council was established to create a mechanism for third party assessment of products, services and processes. In tune with the international best practices, it was envisaged that the Council would ensure that the quality of life of citizens is improved. It was therefore befitting that the Council was organized as an Independent Professional body with representation from the government and the industry, and with its own rules, powers and functions strictly as per the MoA of the Society.

The National Accreditation Board for Certification Bodies (NABCB), a constituent Board of QCI, is the apex national accreditation body which provides accreditation to Certification as well as Inspection Bodies in accordance with ISO Standards, international requirements / guidelines and NABCB Accreditation Criteria. Its objective is to promote acceptance of India's certifications and inspections internationally.

The National Accreditation Board for Testing and Calibration Laboratories (NABL) is another constituent board of QCI that accredits laboratories across scopes. NABL is a Mutual Recognition Arrangement (MRA) signatory member of APAC (Asia Pacific Accreditation Cooperation) and ILAC (International Laboratory Accreditation Cooperation) since 2000.

NABL in the year 2016 achieved the APAC MRA signatory status for Proficiency Testing Providers (PTP) and Reference Material Producers (RMP) schemes along with extension in the existing fields of Testing, Calibration and Medical till December 2020. In October 2019, NABL has also obtained ILAC MRA for Proficiency Testing Providers (PTP) accreditation program (ISO/IEC 17043) in addition to existing Testing including Medical (ISO/IEC 17025 and ISO 15189) and Calibration (ISO/IEC 17025) programs.

QCI's successful record of accomplishment and reputation demonstrates its competence and capabilities across industries. QCI have collaborated in various ambitious projects of the Central and State Governments and have played a critical role in creating an ecosystem for Quality. Some of the highlights of QCI's key projects are listed below:

Projects	Key Highlights
Impact Assessment of PMAY CLSSS Scheme	To evaluate the effectiveness and impact of the Credit Linked Subsidy Scheme (CLSS) under the Pradhan Mantri Awas Yojana (Urban) in addressing the housing needs of various segments of the urban households in India by reaching out to CLSS beneficiaries and non-CLSS beneficiaries.
Dipstick Study for receipt of free grains under Pradhan Mantri Garib Kalyan Anna Yojana (PMGKAY)	- The Ministry of Consumer Affairs engaged QCI to understand the ground reality regarding the implementation of the PMGKAY scheme. This is to be done by undertaking a dipstick study to assess the receipt of free-of-cost food grains under PMGKAY. - The on-ground assessment was conducted in 31 States/UTs covering over 68000 beneficiaries which were approached directly to capture the reach of the scheme. - The study was categorised into four phases and was conducted smoothly and effectively in just over a month.
Public Perception Survey of Delhi Police	The survey aimed to assess public satisfaction with the Delhi Police and identify reasons for this satisfaction, as well as to uncover details about unreported crimes. A total of 15,228 responses were collected, including 2,258 from 1,186 randomly selected households and 1,072 households that filed an FIR after December 31, 2017. Additionally, 15,115 responses were gathered through a telephonic survey targeting individuals who had contacted police helplines.
Socio-Economic Profiling of PM SVANidhi Beneficiaries	- QCI has been appointed by MoHUA to assist in preparing the socio-economic profile of street vendors covered under PM-SVANidhi Scheme, for which QCI has developed a web portal and a mobile application to capture and reflect the data. - More than 22 lakhs street vendor beneficiaries and more than 51 lakhs family members have been profiled. - More than 54 lakhs schemes have been sanctioned to eligible members under eight different schemes. - QCI has now integrated 'Image Recognition Artificial intelligence' with 'SVANidhi Se Samridhi' portal for capturing and validating street vendor photographs during profiling. The functionality was tested/ conducted pilot in two ULBs and now the same is live in all 3514 ULBs.

QCI by virtue of its role in quality and accreditation infrastructure has been receiving a large number of projects from various Central Ministries/Departments and State Governments on **nomination** basis.

4. SCOPE AND OBJECTIVE

The proposed study aims to assess the effectiveness and impact of PM Surya Ghar: Muft Bijli Yojana, implemented by UPNEDA, in providing reliable and sustainable solar energy in Uttar Pradesh and in raising awareness about solar technology. It will evaluate the scheme's implementation across the district of Varanasi by developing an actionable framework to assess its effectiveness. The framework will also identify challenges, analyze areas for improvement, and recommend strategies to enhance its outreach and outcomes in the future.

Broadly, the assessment aims to achieve the following outcomes linked with the specific objectives:

Objectives	Outcomes
1. Evaluate the impact of the scheme in meeting its required objectives.	The scheme's effectiveness in increasing solar rooftop adoption and ensuring reliable electricity. Measure results like solar installations and households served versus demand raised. Evaluate impact on beneficiaries, including energy access and energy security.
2. Cross-sectional Thematic Assessment	The scheme based on cross-sectional themes including beneficiary behavioral change, social inclusion, gender mainstreaming and regional variation.
3. Assess the Economic Empowerment of beneficiaries	The impact of scheme in stimulating household savings, income generation and quality of life.

5. METHODOLOGY

The assessment will utilize a comprehensive approach, combining quantitative interviews and qualitative discussions, to provide a scheme-level evaluation focused on the district of Varanasi. A predefined sample of the beneficiaries will be selected across the district of Varanasi (sample distribution to be shared by UPNEDA) in both urban and rural settlements. Quantitative data will be collected through structured surveys conducted based on practical considerations to provide scheme level analysis. Qualitative insights will be gathered through Focused Group Discussions (FGDs) to be conducted, mostly at block and village level with diverse groups involving implementing stakeholders, opinion makers as well as selected beneficiaries.

Quantitative data will be collected through personal interviews administered to a sample of 1,000 beneficiaries (approx.) across the district. This sample will be proportionally distributed to reflect the rural-to-urban ratio of the beneficiary list, with potential adjustments made based on practical considerations. The 1,000 beneficiaries will be selected using stratified random sampling from a pool of 4,500 beneficiaries (approx.) to ensure a diverse and representative sample. Qualitative insights will be gathered through Focused Group Discussions (FGDs) with key stakeholders, including local functionaries (Key Informants), representatives from power distribution companies (DISCOMs), and empanelled solar vendors. These discussions will provide additional context and depth to the quantitative findings.

The methodology would be finalised in consultation with UPNEDA, ensuring its robustness and obtain findings aligned with the objective of the assessment as detailed below:

Step 1: Framework Development

The project framework for the assessment will be developed with the help of a sector expert to identify key focus areas, stakeholders, and evaluation criteria, including measurable, achievable, relevant, and time-bound KPIs for each assessment method. It will be revised (if required) based on the findings of the pilot study (as detailed in step 4).

Step 2: Design of the Assessment

The proposed methodology for the impact assessment of the scheme will combine quantitative interviews, qualitative discussions, and may also include some outreach strategies. This multifaceted approach will capture diverse perspectives and insights into the implementation, challenges, and impact of the scheme across different beneficiary segments of the district.

Step 3: Sampling Strategy

a) Sample Size: The sample size will be determined based on statistical considerations utilizing stratified random sampling approach. The sample size may be adjusted to account for variations in population density and scheme uptake across the district. The study will encompass a total sample size of 1,000 beneficiaries (approx.) who will be interviewed through various standard means. QCI may require on-ground assistance from the local administration / nodal agency to reach out to beneficiaries.

b) Sample Selection: The representative sample for the Beneficiaries, Key Informant Interviews (KIIs)

and Focused Group Discussions (FDGs) will be designed in such a way that the sample is spread over geographic sub-areas and population sub-groups across the district, covering all the concerned stakeholders.

Step 4: Pilot Study

To ensure the feasibility of our methodology, a pilot study will be conducted in one rural and one urban location within the district. It aims to identify any potential challenges or limitations and make necessary revisions to optimize the suggested assessment design/method.

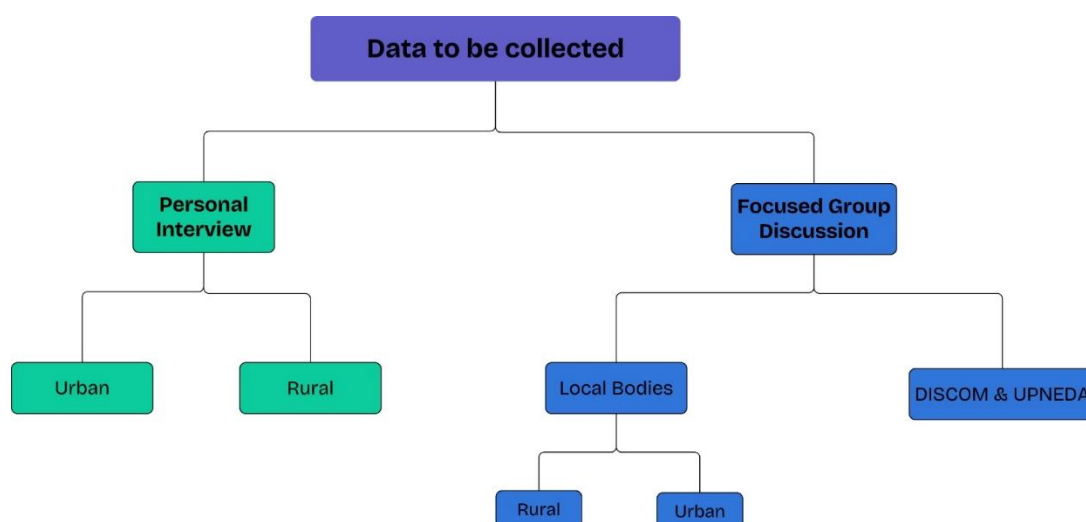


Figure: Snapshot of the Data to be collected from and methods

Step 5: Training of Field Surveyors

To ensure a rigorous assessment process, qualified surveyors with a minimum of four years of professional experience will be selected for deployment. These surveyors will be engaged through the inspection bodies accredited by the National Accreditation Board for Certification Bodies (NABCB), a constituent board of QCI. These inspection bodies adhere to international standards and possess extensive experience in conducting inspections.

Prior to deployment, all the surveyors will undergo comprehensive technical training to acquaint themselves with the assessment framework. This training will ensure a deep understanding of the evaluation criteria and methodologies. Additionally, they will receive ethics training to reinforce the importance of upholding the highest standards of conduct and professionalism throughout the assessment process.

Step 6: Methods to be used for collecting Data

- i. Field Surveys: A subset of the beneficiaries may be surveyed in the field by QCI empaneled surveyors. The survey would be done through a license-based tech platform that QCI will develop to have objectivity and capture required data. The tech platform will be available with all the surveyors to ensure smooth execution of the survey as the data will be synced on a real-time basis (geo-

tagged/geo-stamped) and monitored by the QCI team for all the forms.

- ii. Focus Group Discussions (FGDs): FGDs will be conducted online with key stakeholders with assistance from the local administration/nodal agency. These qualitative discussions will provide deeper insights into the process, outreach efforts, and customer service related to the scheme implementation and impact.
- iii. Data Validation: Data collected shall be validated using the required checklist by the QCI central team. The approach will provide robustness to the collected data by implementing multi-level review of the data.

Step 7: Analyzing the collected data

The quantitative data collected through surveys will be analyzed using statistical techniques, whereas qualitative data from discussions will be analyzed thematically to identify key themes, patterns, and insights relevant to the scheme.

Step 8: Reporting and preparation of final Report

Based on the insights and outcome of the analysis, a report will be prepared summarizing the study findings and recommendations (if any).

6. PROPOSED TIMELINE

Process	Timeline (in weeks)					
	T+2	T+4	T+6	T+8	T+10	T+12
Designing of the Assessment & Sampling Strategy						
Pilot Study						
Training of On-Field Surveyor and Development of Survey Tech -Based Application						
Roll out of the Assessment						
Data Review & Analysis						
Preparation and Submission of Draft Report						
Preparation and Submission of Final Report						

Note:

- 1) 'T' here is the date of issuance of work order by the UPNEDA and acceptance by QCI.
- 2) In this proposed timeline, mandatory holidays are excluded, indicating non-working days.

7. FINANCIALS

S. No.	Particulars	No. of Unit	Unit Cost (in INR)	Cost (in INR)
1	Project Implementation Team			
1.1)	Technical Expert (for 0.5 months)	1	300000	₹ 1,50,000
1.2)	QCI Team (for 3 months)	6	75000	₹ 13,50,000
2	Survey Team *			
2.1)	On Ground Team (for 1 month)	100	7000	₹ 7,00,000
3	Technology Support			
3.1)	Out Reach Activities	1000	4.8	₹ 4,800
3.2)	IT Cost - App used for on Field Survey	Lumpsum		₹ 45750
4	Data Analysis & Visualization	Lumpsum		₹ 1,25,000
5	Report Designing and Printing	Lumpsum		₹ 1,75,000
	Sub Total			₹ 25,50,550
	Admin Cost		12%	₹ 3,06,066
	Total (with Admin Cost)			₹ 28,56,616
Plus, GST @18%				

*Data will be collected from 1,000 beneficiaries(approx.) using a questionnaire comprising of 15 to 20 key questions.

Note:

- GST to be charged as per the norms (18% on the Grand Total).
- The financial calculation provided is limited to the scope described above (section 4).
- Any changes to the scope will incur additional costs, which will be billed separately and is not included in the proposed financials.
- In case of any delay in the on-field assessment due to concerns pertaining to change of a procedure, lack of support by the engaged stakeholders etc., that results into additional man-day will be considered as extra cost will be charged to UPNEDA separately.

8. TERMS OF PAYMENT

The table below shows the terms of payment to be made for the project:

Milestone or Reporting Deliverable	Project Amount to be disbursed to QCI
Award of Work-Order	40%
Completion of Data Collection	40%
Submission of Final Report	20%
Total	100%

9. SUPPORT REQUIRED

1. Appoint a Single Point of Contact (SPOC) for all the future communication and support.
2. UPNEDA will provide the complete Discom-wise dataset for Varanasi, including information on PMSGMBY beneficiaries (name, phone number, address, beneficiary category, etc.), to facilitate the study.
3. UPNEDA to provide the complete data concerning the demands raised and the sanctioned applications, including the list of beneficiaries, for the assessment study. Issuance of official authorization letter from the UPNEDA to QCI endorsing the survey task and providing necessary permissions and support for data collection activities, wherever required.
4. Provide all the necessary support to organize Focus Group Discussions (FGDs).
5. UPNEDA to share any other information/assistance as requested by QCI for the project execution and completion.

GENERAL TERMS

- **AMENDMENT OF AGREEMENT:** No amendment or modification of the agreement authorizing this project shall be valid unless it is made in writing jointly by both parties or by their authorized representatives and specially stating the reason for amendment to this agreement. The modification/changes shall be effective from the date on which they are made/executed unless otherwise agreed to.
- **FORCE MAJEURE:** The parties shall ensure due compliance with the terms of this agreement. However, no party shall be liable for any claim for any loss or damage whatsoever arising out of failure to carry out the terms of the agreement to the extent such a failure is due to force majeure events such as fire, rebellion, mutiny, civil commotion, riots, strike, lockout, force of nature, accident, act of God, or any other reason beyond the control of concerned party.
- **ARBITRATION:** The disputes, if arising, shall be settled by arbitration with sole arbitrator (to be jointly appointed in 30 days) in accordance with the procedure prescribed in the Indian Arbitration and Conciliation Act, 1996 or any statutory modification or re-enactment thereof and the rules made thereunder and for the time being in force. Proceedings shall, unless otherwise agreed by the parties, to be held in Delhi, India. Governing law will be laws of India. Language will be English.
- **NOTICE:** All notices permitted or required to be given under the provisions of this agreement shall be deemed to be sufficiently given if written and delivered personally or sent to the other party by telefax or certified or registered air mail or email and addressed to the registered office of the other party as follows or to such other party address as the party may give notice in writing. Unless otherwise provided herein all such notices shall be deemed to have been given as of the date they are personally delivered or transmitted by telefax or email or deposited in the mail.
- **TERMINATION OF AGREEMENT:** Either party shall have the right to terminate the agreement at any time by giving thirty days' notice in writing in advance indicating reasons for same to the other party. The reasons however, will need to be justified and in case the termination leads to delay/pause/abortion of the project, the Authority reserves the right to not make any payments further to the date of project termination as indicated by either party.